

Executive Summary

Title:

TikTok Project: Predicting Claim vs. Opinion

Subtitle:

Random Forest vs XGBoost Models

Project Overview:

An exploratory data analysis (EDA), data cleaning, data transformation, data splitting, model/s training, and testing were conducted to build a model that predicts whether a video is a claim or an opinion. The champion model (the random forest model) shows that the video transcription text length is the most predictive feature, followed by author-ban-status-active. The results make sense as claims usually require more explanation, evidence, and context, which translates into longer transcriptions. The authors who report claims are most likely to be heavily scrutinized and placed under banned or under review status, making active status highly predictive, as it strongly signals the absence of a claim. The absence of the claim here is valuable as it allows the model to reveal patterns in the data.

Key Insights:

-Model Performance: The Random Forest model achieved an accuracy score of 69%. However, the accuracy score is not our target in this analysis, as we need our model to catch as many claims as possible; therefore, we prioritize the recall score. Since the random forest yields a higher recall score (76%) in the validation set compared to the XGBoost model (73%), the random forest was the champion model. When evaluated against the test set, the random forest (the champion model) successfully identified 1803 claims and 1411 opinions, yet it also misclassified 989 opinions as claims (false positives) and 568 claims as opinions (false negatives).

-Video Transcription Text Length: The model shows that video Transcription Text Length was the most predictive feature, which means more analysis should be implemented to reveal the underlying patterns that distinguish claims from opinions.

- Video duration in seconds, followed by verified status, has the minimal association with the claim status.

Variables Test:

Dependent variable: claim_status (categorical)

Independent variables: video_duration_sec, video_transcription_text_length, verified status, author_ban_status.

Next Steps:

-More data should be collected related to audio features like the emotional tone of the voice, the background music, or visual features extracted from the video frame, such as objects or text within the video frame.

-Data should also be extracted from the feature that the model connotes as the most predictive feature (video transcription text length), such as the common words among videos flagged as claims. This could be done using n-grams or sentiment analysis.